

Biaxial Mooney-Rivlin coefficient of silicone sheet by additive manufacturing

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Abstract

This study evaluates the biaxial Mooney-Rivlin coefficient for silicone sheets made by additive manufacturing (AM) for custom wearable assistive devices and soft robotics. The extrusion-based layer-by-layer technique is applied for AM of moisture cured silicone sheets. A progressive cavity pump controls the precise, continuous flow of high viscosity silicone for low void AM of the cruciform shaped specimen. A biaxial tensile test apparatus with four independent linear actuators was built for testing the cruciform specimens. Digital Image Correlation method was utilized to calculate the distribution of strain. This study show the value of the biaxial Mooney-Rivlin coefficient for the AM silicone sheet is independent of the thickness and in the range of 114.7 kPa to 132.1 kPa for C_{10} and -17.9 kPa to -9.04 kPa for C_{01} . The geometry of the specimen is shown to affect the biaxial quality of the specimen.

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1. Introduction

Advancements in additive manufacturing (AM) have enabled the personalized design and broad applications of custom products. One area with great technical challenges and potential for growth is the individualized wearable assistive devices, which require soft and stretchable components that fit a person's needs. The other area is for soft robotics, particularly the grippers, pneumatic actuator [1], flexible and stretchable sensors [2]. Silicone elastomers, denoted as silicone, has the mechanical properties ideal for custom components of these applications [1]. Traditionally, molding is the fabrication process for silicone. However, the time and cost for mold making have inhibited the adoption of custom silicone parts. Molding also has limited capability on the shape, such as the internal cavity and thin wall. The AM of silicone opens opportunities for new design and requires the understanding of mechanical properties under loading.

The extrusion-based silicone AM in this study is a line-by-line and layer-by-layer process [3]. The AM process

parameters such as layer height, width, and line direction all affect mechanical properties of the silicone sheet in different orientations. The layer deposition in extrusion-based AM process leads to anisotropy which needed to be evaluated through biaxial testing.

Biaxial tensile test elongates a thin specimen in two perpendicular directions and measures the deformation. Based on measured strain and nominal stress, the fundamental material properties can be calculated. The biaxial testing of thin hyperelastic material has been performed on a cruciform specimen [4] using the Digital Image Correlation (DIC) method to track movements of powder sparsely spread on the surface of the specimen. The deformation between two adjacent particles were used to calculate the strain. The distribution of strain was mapped on the stretched specimen to find the biaxial elastic moduli. Seibert [4] presented four shapes of the thin hyperelastic specimens, which are the foundation of the AM silicone specimen in this study. Another study by Lowell et al. provided the biaxial fixture design which

utilizes two linear axes to pull the cruciform specimen by a constant force in all four sides [5].

In this study, thin cruciform silicone sheet specimens are made by extrusion-based AM. One of the goals of this study is to provide guidelines on specimen geometry design for biaxial testing of thin silicone sheet made by AM. The other goal is to evaluate the biaxial coefficient of the silicone using Mooney-Rivlin criterion under effect of AM process parameters.

2. Materials and Methods

2.1. Silicone material

The moisture cured silicone (Dow Corning 737) was chosen as to make the specimen by AM. This silicone has a high viscosity of 45 Pa·s at 27°C and quick curing time of 14 min to maintain the shape during AM. The maximum elongation it can achieve is 300% with tensile strength of 1206 kPa [1]. The cured product will have 33 Shore A Hardness [6].

2.2. AM of silicone biaxial test specimens

The setup for silicone AM of cruciform specimen is shown in Fig.1. The machine is based on a 3D-printer (TAZ 5 by Lulzbot, Loveland, Colorado) and has three linear axes (X, Y, and Z). The line-by-line building of the specimen in a layer was conducted in the X-Y plane. The layer is built in the Z direction. The extrusion of silicone is done utilizing a progressive cavity pump (PCP) (Model Preeflow eco-PEN 450 by ViscoTec, Töging am Inn, Germany). This PCP allows for a high precision extrusion up to 0.004 ml with more than 99% repeated accuracy [7]. Next to the PCP are syringe tubes that stores the silicone, which feeds to a manifold that supplies to the PCP.

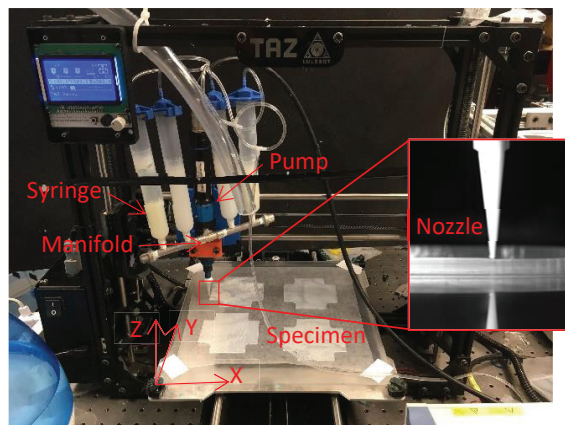


Fig. 1. Setup for AM of silicone.

Fig. 1 shows four samples on the build plate for AM. The symmetrical cruciform shape allows for biaxial-planar loading to be applied to the specimen. The load is applied at the clamped region of the specimen specified as the tab zone [8],

as shown in green box with width A and depth D in Fig. 2. In this study, A was 50 mm. The clamping jaw was 55 mm wide (allowing 2.5 mm on each side for the expansion of the specimen during clamping). The depth of the tab zone D was 13 mm while the jaw depth was 16.2 mm (allowing 3.2 mm for the expansion of the specimen during clamping).

The fillet radius R was adjacent to the tab zone. In Fig. 2, the specimens have an overall length $L = A + 2R + 2D$. In this study, $R = 5$ mm and $L = 86$ mm. The gauge zone [9], as shown in the yellow box in Fig. 2, is a region at the center of the specimen which will be observable to a digital camera for strain measurement. The gauge zone has its four corners touch the fillet curve and is a square with length G with $G = A + 2(R - (R\sqrt{2})/2)$. In Fig. 2, $G = 52.9$ mm. Another key geometrical parameter is the thickness B of the specimen. A thicker and more number of layers will increase the cross-sectional area and will affect the surface compression between layers during AM and may change the mechanical properties. The gauge zone side length G and B are used to determine the cross-sectional area required to determine the stress of the specimen.

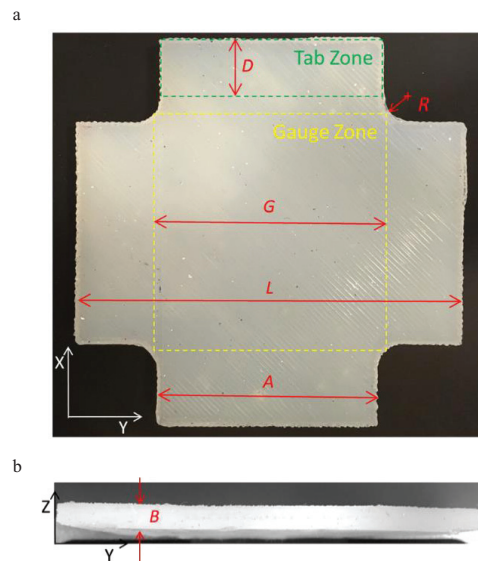


Fig. 2. (a) Cruciform specimen and geometrical parameters and (b) specimen midsection cutaway.

The fillet radius will also affect the stress and strain distributions which yield varying biaxial quality of the specimen. In this study, six specimens with values of R and B along with G are listed in Table 1.

Table 1. Geometrical parameters B , R and G for six specimens.

Specimen	Thickness B (mm)	Fillet Radius R (mm)	Gauge Zone Length G (mm)
1	1.0	5.0	52.9
2	2.0	5.0	52.9
3	5.0	5.0	52.9
4	8.0	5.0	52.9
5	1.0	14.5	58.5
6	1.0	18.5	60.8

Minimizing voids while maintaining geometrical accuracy is paramount. To mitigate specimen defects due to voids and maintain geometrical accuracy, the AM parameters were adopted from a study using the same silicone material for low void AM of silicone [1]. The nozzle diameter used was 0.41 mm, the layer height was 0.20 mm, the extrusion volumetric flowrate was 0.12 ml/min, the speed of nozzle was 20.0 mm/s, and the distance between infill was 0.42 mm during AM. Midsection cutaway of the printed cross-section can be seen in Fig. 3 for $B = 2, 3, 5$ and 8 mm to demonstrate that there was essentially no void. The cross-section views of four specimens in Fig. 3 show uneven surfaces which could cause stress concentrations.

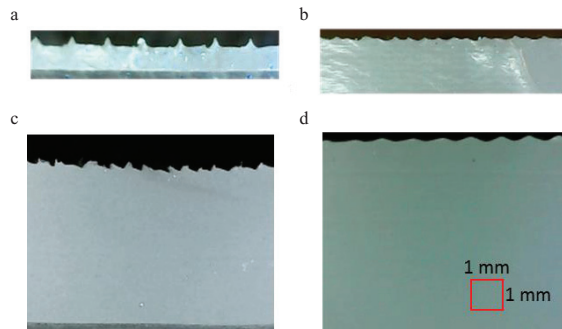


Fig. 3. Midsection cutaway for (a) 2 mm, (b) 3 mm, (c) 5 mm and (d) 8 mm specimen thickness showing no void and uneven specimen surface.

2.3. Experimental Setup

The biaxial tensile test apparatus, as shown in Fig. 4, consists of 4 linear actuators, each fitted with a stepper motor and ball screw (Model HS-02 by Dynamic Solution, Irvine, California). Two linear actuators are placed on axis X, whereas the other two linear actuators are placed on axis Y. In each axis, two linear actuators are moving synchronously in opposite directions. One-bolt clamps were used to hold specimen on each edge to ensure no movement with respect to the linear actuators. The specimen was elevated between four clamps and its center was kept in the middle of the setup during test. All four linear actuators have independent movement, which allows for various trajectory of loading and movement scenarios. The speed of each linear actuator was set at a minimum of 2 mm/s to minimize the viscoelastic effect and allow for a quasi-static deformation assumption [10].

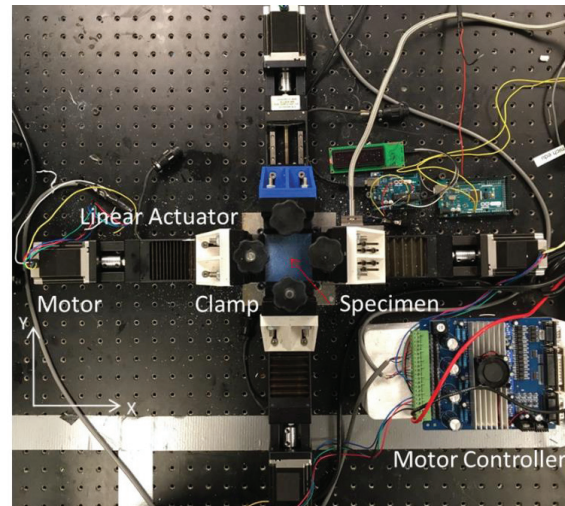


Fig. 4. Biaxial test setup for silicone sheet specimen covered with powder.

2.4. Strain Measurement

A digital camera (not shown on Fig. 4) was placed on the top of the specimen to record the movement of particles on the top of the silicone sheet during the biaxial tensile test. NCORR™, an open sourced Mathworks™ software, was utilized for the DIC analysis of the deformation on an object's surface [11]. The strain distribution on the specimen under biaxial tensile test can be determined by acquiring the deformation map [12] in the gauge zone from comparing consecutive images. Each image holds several subsets with each subset comprised of several control points. The control points hold the greyscale information of the subset. By comparing correlation values of an array of subsets between several consecutive images, the movement and deformation of the specimen are determined through the updated values of u and v (the vector field on X- and Y-direction, respectively) and their derivatives $\frac{\partial u}{\partial x}$, $\frac{\partial v}{\partial x}$, $\frac{\partial u}{\partial y}$, and $\frac{\partial v}{\partial y}$ [11]. In this study, the subset has a radius of 35 pixels and the control points have a spacing of 6 pixels. The iteration ended when a threshold difference of less than 0.001 correlation coefficient value between each image or 20 iterations is reached.

The DIC parameters are determined to optimize computing time and resolution of the deformation map. Larger subset size may increase accuracy of DIC by sacrificing the computing time. The strain radius in DIC used to calculate strain is 15 pixels. Smaller strain radius will introduce noise to the calculation, while larger strain radius may not have accurate strain prediction [11]. Through the acquired deformation, the principal strain in X and Y directions and the shear strain can be determined.

2.5. Specimen Shape Quality Criterion

Aside from easy reproducibility, a qualified specimen will need to exhibit homogenous stress and strain in the gauge zone of the specimen. Along with strain variability within the gauge zone, the homogenous stress and strain properties will determine the quality of the specimen shape for biaxial tensile test [13]. A methodology based on guidelines provided by Seibert [4] for biaxial tensile test quality criterion is developed for silicone based AM cruciform specimen. Distributions of the principal strain in X and Y directions are first acquired. The absolute difference between X and Y principal strains over the midpoint centerline $A-B$ (as shown in Fig. 5) are then examined.

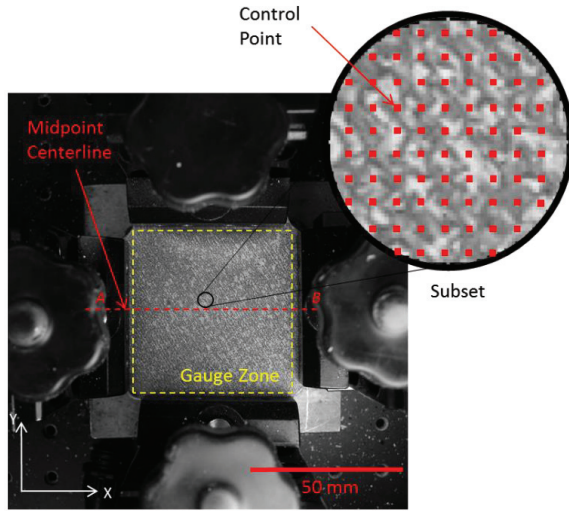


Fig. 5. Evaluated midpoint centerline with a subset magnification showing distribution of control points in the subset.

When the difference in X and Y principal strain values on the centerline is equal or close to zero, the specimen is said to exhibit a high biaxial quality. Although it is impossible for all the region to have zero difference between each principal strain, the specimen with the lowest difference in principal strain value within the centerline is preferred. The principal strain error τ [4] can be measured using the following test function:

$$\tau(x, y) = |E_{xx}(x, y) - E_{yy}(x, y)| \quad (1)$$

where E_{xx} and E_{yy} are the strain in the X and Y direction, respectively. At the centroid [4] of the specimen, $\tau = 0$ as the principal strains are equal: $E_{xx}^c \equiv E_{yy}^c$.

Another indicator used for biaxial quality is the standard deviation of the $A-B$ midpoint centerline s . To introduce this factor to the biaxial quality, the sum of variance for strain data within 15 pixels away from and perpendicular to the midpoint centerline is measured. The distance from the midpoint

centerline is chosen based on the strain used for the DIC strain measurement.

Apart from the standard deviation, another aspect of biaxial quality is determined by evaluating the difference between the maximum principal strain value E_{xx}^{max} with the strain value at the centroid E_{xx}^c . The relationship is defined as the degree of efficiency η [4]:

$$\eta = E_{xx}^c / E_{xx}^{max} \quad (2)$$

Both degree of efficiency η and standard deviation σ_c will be contributing factors to the quality criterion Q [4]:

$$Q = \eta / s \quad (3)$$

Along with the principal strain error, the quality criterion is one of the main determinants of biaxial quality of the specimen. A high value of Q is desired for a specimen to provide a large area of biaxial deformation while having a relatively small amount of strain variation.

3. Results and Discussion

The biaxial tensile test setup for Specimen #6 in Table 1 with $R = 18.5$ mm and $B = 1.0$ mm is shown in Fig. 5. The distribution of the control point is shown on a close-up image of one of the subsets. The total control point for each subset was 78 pixels.

Shown in Fig. 6 are the strain distributions for E_{xx} and E_{yy} for specimen #6. The strain value $E_{xx}^{max} = 0.227$ and $E_{xx}^c = 0.121$, which resulted $\eta = 0.53$. The maximum τ was determined by the maximum difference between E_{xx} and E_{yy} on the midpoint centerline which yielded $\tau = 0.124$. The strain distribution indicated that the area close to the centroid is exhibiting biaxial deformation since the difference between E_{xx} and E_{yy} is low. This is further demonstrated by comparing the principal strain error between each fillet radius as shown in Fig. 7. The principal strain error for varying thickness is shown in Fig. 8.

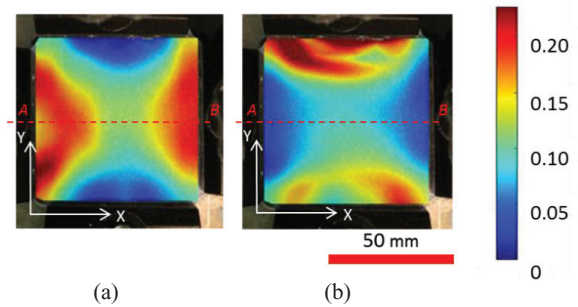
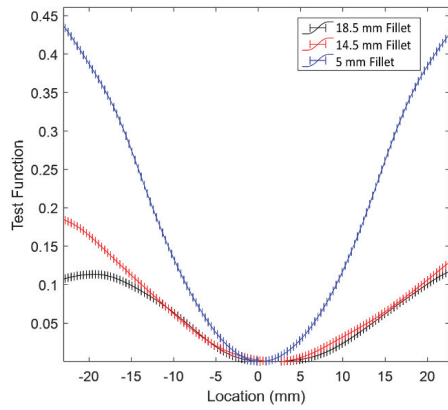
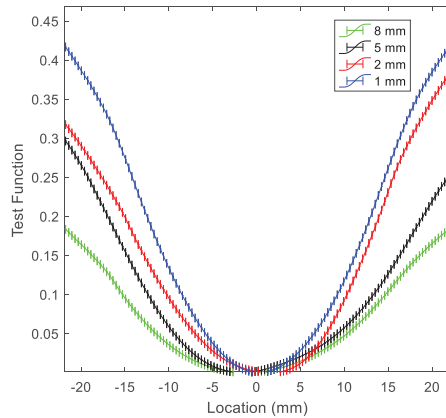


Fig. 6. (a) E_{xx} and (b) E_{yy} for 18.5 mm fillet radius, 1 mm thick specimen acquired from DIC data.

Fig. 7. Principal strain error value τ for various fillet radius.Fig. 8. Principal strain error value τ for various thickness

Specimens with smaller R tend to show a larger shift in τ whereas thinner specimen tends to show a smaller shift. For every sample, the change in τ starts slowly from the centroid where $E_{xx} = E_{yy}$ while gradually increases when it is moving toward the edge of the specimen. The large value of τ at the edges is due to the specimen exhibiting uniaxial deformation as principal strain in the Y direction is at its minimum. Complete σ_c and Q values can be seen in Table 2.

Table 2. Changes in biaxial quality due to thickness and fillet radius change.

Thickness B (mm)	Fillet Radius R (mm)	Standard Deviation s	Scaled Degree of Efficiency η	Principle Strain at centroid E_{xx}^c	Quality Factor Q
8	5	0.018	0.45	0.197	25
5	5	0.031	0.38	0.221	12.25
2	5	0.052	0.74	0.588	14.23
1	5	0.483	0.63	0.466	1.30
1	14.5	0.092	0.45	0.096	4.89
1	18.5	0.028	0.53	0.121	18.92

The significant reduction in Q resulted from surface imperfection due to AM process, which results in higher strain variability for thinner specimen. The proof of this imperfection can be examined in the cross-section photograph of the specimen as shown in Fig. 3. The imperfection is most likely to be caused by the presence of voids at the top layer surface of the biaxial specimen. As more layers are printed, these voids will gradually decrease due to material pressed from the upper layer which caused the bottom layer to expand in its width thus closing off the voids. The voids caused the strength between adjacent layers to be reduced and caused the strain distribution to be less uniform for thinner material when compared to a thicker one. In addition to the difference in thickness, the difference in fillet radius also contributes to the quality criterion. Based on comparison of specimens #1, #5 and #6, as the radius gets smaller, σ_c gets larger and Q will be reduced. The larger σ_c value indicates that for a smaller R , the region adjacent to the centerline tends to experience less uniform strain distribution. This is due to the presence of higher stress concentration for smaller R .

The biaxial properties of the specimen can be determined by utilizing Mooney-Rivlin criterion for hyperelastic material [14]. Through this criterion, the strain energy density W [10] is:

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3) \quad (4)$$

where C_{10} and C_{01} are the material constants [10]. I_1 and I_2 are the strain invariants [10] which are functions of the three-principal stress with λ_1, λ_2 and λ_3 correspond to X, Y and Z axis principal stretches respectively [10]:

$$I_1 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2 \quad (5)$$

$$I_2 = (\lambda_1\lambda_2)^2 + (\lambda_2\lambda_3)^2 + (\lambda_1\lambda_3)^2 \quad (6)$$

$$I_3 = \lambda_1^2\lambda_2^2\lambda_3^2 \quad (7)$$

For an isotropic and incompressible material, W can be expressed as a function of the principal stretches λ and stress σ [10]:

$$\sigma_1 - \sigma_3 = \lambda_1(\partial W / \partial \lambda_1) - \lambda_3(\partial W / \partial \lambda_3) \quad (8)$$

Due to an assumption of incompressible materials, the third invariant I_3 is equal to one [10]. Therefore, the strain-energy function will depend on two invariants. Following this procedure, the Mooney-Rivlin model can be modified for biaxial tension. Substituting $\sigma_2 = \sigma_1$, $\sigma_3 = 0$, $\lambda_2 = \lambda_1$, $\lambda_3 = \lambda_1^{-1}/\sqrt{2}$ will yield the following equation:

$$\sigma_1 = 2(\lambda_1^2 - \lambda_1^{-4})(C_{10} + C_{01}\lambda_1^2) \quad (9)$$

The global nominal stress [15] can be acquired by dividing the force F given to each arm of the specimen with the cross-section of the stressed area. Since the applied load is equal for both axes, the stress on each axis is defined as:

$$\sigma_x = \sigma_y = \sigma = \frac{F}{GB} \quad (10)$$

Based on the measurement, the stress and strain data are then fitted through ANSYS Curve Fitting Tool® to determine the Mooney-Rivlin coefficients with the smallest residual value. The coefficients that yields the minimum objective functions are shown in Table 3.

Table 3. Mooney-Rivlin coefficients

Thickness <i>B</i> (mm)	Fillet <i>R</i> (mm)	C_{10} (kPa)	C_{01} (kPa)	<i>RMSE</i> (kPa)
1	18.5	132.1	−12.8	19.0
1	14.5	126.4	−14.6	23.9
1	5	114.7	−16.1	2.54
2	5	125.9	−17.9	8.68
5	5	118.9	−10.8	17.6
8	5	114.8	−9.04	7.69

Table 3 indicates that larger fillet tends to have a larger C_{10} value. C_{10} represents the elastic behavior of the material and an increase in C_{10} indicates a stiffer material [16]. The higher value in C_{10} can be caused by the even distribution of stress on larger fillet.

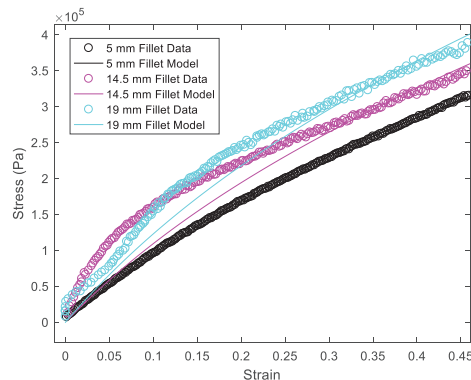


Fig. 9. Mooney-Rivlin model and data comparison for various fillet size

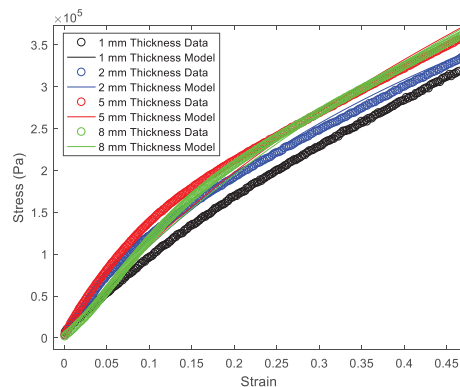


Fig. 10. Mooney-Rivlin model and data comparison for various thickness

There was no trend identified on the coefficients by modifying the thickness of the specimen although the model indicates that the thicker material corresponds to a stiffer material as shown on Fig. 10. Both thickness and fillet size of the specimen effects the inflection point where the stress-strain relationship becomes linear. Non-linearity and location of inflection point effects the error of the model as shown by the root-mean-squared error *RSME*. *RSME* value indicates that the fitness of the data to the 2-order Mooney-Rivlin model. Higher *RSME* value indicates that the current order of the model is less compatible with the data therefore, more coefficient is needed to fit the data. C_{01} value indicates the non-linearity of the stress-strain curve [16]. An increase in C_{01} specifies a less linear curve. There was no trend in C_{01} and the values are in range of −17.9 kPa to −9.04 kPa. The range in C_{01} can be caused by the non-uniform distribution of strain at the gauge zone which contributes to the measurement error. The value of the Mooney-Rivlin coefficients indicates that the AM silicone sheet is more flexible when compared to carbon black reinforced natural rubber due to the significant difference in C_{10} value where $C_{10} = 333.9$ kPa [17] for carbon black reinforced natural rubber. The AM silicone sheet also has a less linear stress-strain relationship when the C_{01} value is compared where $C_{01} = -3.37$ kPa for carbon black reinforced natural rubber.

4. Conclusions

This study provided a guide to biaxial test and geometrical effects of cruciform-shaped silicone sheet test specimen made by AM undergoing an equibiaxial load. Results demonstrated that AM can fabricate flexible silicone sheets for various potential applications. Independent motor used for the setup will allow for alteration of deformation mode. A specimen that exhibits a large biaxial region while minimizing variation and difference between maximum principal strain and minimum principal strain is desired. The DIC method generated a quantifiable strain distribution for each specimen. Through the determination of the Q , thicker specimen with the largest fillet radius exhibited a desirable biaxial quality. The result showed that the Mooney-Rivlin coefficient acquired through various specimen appear to be within a range of values for 0 to 100% strain. The C_{10} value was at a range of 114.7 kPa to 132.1 kPa whereas the C_{01} value range from −17.9 kPa to −9.04 kPa. The AM silicone sheet is shown to be more elastic due to the smaller C_{10} value compared with carbon black reinforced natural rubber, which is a desirable characteristic for wearable material aside from the nonsensitizing and nonirritating requirement [18]. Larger fillet and thicker specimen should be utilized for AM silicone sheet specimen for biaxial tensile test since it exhibits the highest biaxial quality. Repeatability is a future work for both measurement and AM.

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